

Table 1: Residual-objective ablation under the Double Sphere model. All modes share the split, initialization, fixed backend frame-board inputs, and frozen holdout observations within each sequence. Aggregates are sequence-level macro-averages over three independent captures; best results are in bold.

| <i>(a) Holdout image-plane accuracy</i>                |                            |               |               |               |
|--------------------------------------------------------|----------------------------|---------------|---------------|---------------|
| Residual                                               | Seq. A                     | Seq. B        | Seq. C        | Mean          |
|                                                        | Reprojection RMSE [px] ↓   |               |               |               |
| Pixel                                                  | 0.5193                     | 0.5042        | 0.5269        | 0.5168        |
| Spherical                                              | 0.5195                     | <b>0.5035</b> | 0.5269        | 0.5167        |
| Hybrid                                                 | <b>0.5191</b>              | 0.5042        | <b>0.5266</b> | <b>0.5166</b> |
| Kalibr (DS)                                            | 0.6284                     | 0.5666        | 0.6386        | 0.6112        |
| <i>(b) View-angle-conditioned ray accuracy</i>         |                            |               |               |               |
| Residual                                               | 0–30°                      | 30–50°        | ≥ 50°         | Overall       |
|                                                        | True angular RMSE [mrad] ↓ |               |               |               |
| Pixel                                                  | 0.3204                     | 0.3493        | 0.4117        | 0.3488        |
| Spherical                                              | 0.3205                     | <b>0.3489</b> | <b>0.4107</b> | <b>0.3486</b> |
| Hybrid                                                 | <b>0.3202</b>              | 0.3491        | 0.4114        | 0.3486        |
| <i>(c) Cross-sequence geometric stability and cost</i> |                            |               |               |               |
| Residual                                               | Overall                    | Outer         | Cross-seq.    | BA time       |
|                                                        | RMSE [px]                  | RMSE [px]     | ray RMS [deg] | [s]           |
| Pixel                                                  | 0.5168                     | <b>0.1255</b> | 0.2334        | <b>1.0</b>    |
| Spherical                                              | 0.5167                     | 0.1275        | <b>0.2237</b> | 10.3          |
| Hybrid                                                 | <b>0.5166</b>              | 0.1277        | 0.2281        | 12.6          |

*Protocol.* Polar bins are assigned once using the shared initialization camera. Cross-seq. ray RMS is the pairwise ray-curve disagreement over 1,422 shared image locations (4,266 ray pairs). BA time is the mean incremental-optimization runtime. All metrics are lower-is-better.